C. Industrial Nim

time limit per test: 2 seconds
memory limit per test: 64 megabytes

There are n stone quarries in Petrograd.

Each quarry owns m_i dumpers ($1 \le i \le n$). It is known that the first dumper of the i-th quarry has x_i stones in it, the second dumper has $x_i + 1$ stones in it, the third has $x_i + 2$, and the m_i-th dumper (the last for the i-th quarry) has $x_i + m_i - 1$ stones in it.

Two oligarchs play a well-known game Nim. Players take turns removing stones from dumpers. On each turn, a player can select any dumper and remove any non-zero amount of stones from it. The player who cannot take a stone loses.

Your task is to find out which oligarch will win, provided that both of them play optimally. The oligarchs asked you not to reveal their names. So, let's call the one who takes the first stone «`tolik`» and the other one «`bolik`».

Input

The first line of the input contains one integer number n ($1 \le n \le 10^5$) — the amount of quarries. Then there follow n lines, each of them contains two space-separated integers x_i and m_i ($1 \le x_i, m_i \le 10^{16}$) — the amount of stones in the first dumper of the i-th quarry and the number of dumpers at the i-th quarry.

Output

Output «`tolik`» if the oligarch who takes a stone first wins, and «`bolik`» otherwise.

Examples

input Copy

```
2
2 1
3 2
```

output Copy

```
tolik
```

input Copy

```
4
1 1
1 1
1 1
1 1
```

output Copy

```
bolik
```

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 15

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
370258137	Apr/08/2026 12:43	Accepted
370257959	Apr/08/2026 12:42	Compilation error

→ Problem tags

games *2000

No tag edit access

→ Contest materials

- Announcement
- Tutorial #1 (ru)
- Tutorial #2 (en)

Codeforces (c) Copyright 2010-2026 Mike Mirzayanov
The only programming contests Web 2.0 platform
Server time: Apr/08/2026 16:47:08 UTC+7 (n2).
Desktop version, switch to mobile version.
Privacy Policy | Terms and Conditions

Supported by